Lesson 3.4: Ocean Acidification

Location: Classroom

Grade Level: High School

Time: Two class sessions of 50 minutes

Driving Questions
- How does increased carbon dioxide in the atmosphere cause ocean acidification?
- How does ocean acidification affect the ocean's food web?

Lesson Synopsis
During this lesson, students investigate how rising atmospheric carbon dioxide levels are a catalyst that can lead to a change in the ocean's chemistry, in particular, increasing the rate of a process known as ocean acidification. They then explore how this rate of change can impact marine organisms made of calcium carbonate shells and eventually the ocean food web. They also explore how these changes could affect human reliance on ocean resources, such as fisheries and coastal livelihoods.

On day one, students connect their understanding of how carbon dioxide in the atmosphere dissolves in seawater to form carbonic acid, lowering the pH of the ocean's water and causing it to become more acidic. They examine how this pH change can affect organisms, particularly those with calcium carbonate shells. Students create a model to demonstrate why they think this impact is happening.

On day two, students investigate this phenomenon further by exploring a SageModeler model that demonstrates how an increase of human-created carbon dioxide impacts the ocean's pH, and ultimately the organisms that live in the ocean, cascading through the entire food web.

Students finally reflect on how they may modify their system models to incorporate their newfound understanding of ocean acidification and its impacts on food webs.

What students will figure out:
- Rising atmospheric CO_2 dissolves into ocean water, forming carbonic acid that lowers the ocean's pH and increases acidity. This process is known as ocean acidification.
- Increased acidity weakens marine organisms with calcium carbonate shells or skeletons, altering interconnected relationships in the ocean food web.
- Human dependence on ocean resources, such as fisheries, food, and coastal economies, is affected by these chemical and ecological changes.

LEARNING OUTCOMES

By the end of this investigation, students will be able to…	You can assess this using…
1. Explain how an increase in carbon dioxide levels contributes to the process of ocean acidification.	Group discussion during the *Navigate* section
2. Construct a claim supported by evidence and reasoning about the long-term effects of ocean acidification on ecosystem stability.	Group discussion during the *Investigate* section
3. Plan how they will update their SageModeler models with their newfound information about ocean acidification and food web impacts.	Group discussing in the *Navigate* section
4. Reflect on how changes to the ocean's chemistry can affect ocean food webs and human access to its resources.	Group reflection in the *Problematize* section

Building towards NGSS and AI Literacy:

- **HS-ESS3-6:** Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. In this lesson, students will construct a model based on local climate change and how human activity contributes to positive and negative effects in the environment.

Lesson Overview (2 sessions)
This lesson includes a shared teacher-led introduction with Lesson 3.3 Sea Level Rise and Coastal Erosion and then transitions into the student-led portions.

Teacher Led Portion Day 1: What do We Want to Learn More About? (10 minutes)

Section	Time	What Happens
Navigate	10 minutes	This is a teacher-led section of the program that gets the class started for Lessons 3.3 or 3.4. Students begin by reflecting back on how increased levels of carbon dioxide emissions lead to changing weather and how that can lead to an increased rate of global warming.

Student Led Lesson Day 1: Exploring the Ocean Acidification (40 minutes)

Section	Time	What Happens
Navigate	5 minutes	Students begin the investigation by exploring the impacts ocean acidification can have by observing an image of a shell that has been placed in vinegar. Students make observations and prepare to further explore this phenomenon by setting up their own experiment.
Investigate	25 minutes	To investigate this phenomenon, students set up an experiment that they will observe over the course of the two lessons. In their experiment, two containers, one with a basic solution (soap water) and another with an acidic solution (vinegar), are set up with a shell placed inside them. These two containers represent two scenarios of ocean pH conditions. Students take a moment to make their initial observations as to what is happening in each container and draw a model that relays any processes or reactions that are taking place. Finally, they make predictions on what they think will happen to the shells between the set-up time and the next lesson, where they revisit their container.
Navigate	5	As a class, students come together to explain what they did and

	minutes	what they observed during day one of their observations. Students also share initial ideas of how they can build their understanding of carbon dioxide's role within their investigation.
Problematize	5 minutes	Students reflect on how they can continue to learn more about ocean acidification.

Student Led Lesson Day 2: Exploring Ocean Acidification's Impact on the Food Web (50 minutes)

Section	**Time**	**What Happens**
Navigate	10 minutes	Students begin the investigation by first watching a video to further learn about ocean acidification and then revisiting the image of a shell that has been placed in vinegar. Building off of their understanding from day one, they share new insights they have.
Investigate	25 minutes	On the second day of their exploration, students start by bringing their experiment from day one back out to make additional observations. Students work as a team to make observations on how their experiments have changed since the last lesson and see if their predictions were supported. They also share why they think they received the result that they did. Students then delve into a pre-set SageModeler model where they get to further explore the impacts that human-linked carbon dioxide emissions have on the ocean ecosystem by increasing the rate of ocean acidification and ultimately impacting the values their communities place on Southern California's ocean.
Navigate	10 minutes	Once students have gained knowledge about what causes ocean acidification and how it can impact coastal communities in Southern California, they work together to create a plan to update their SageModeler model from Module 2 with their newfound knowledge.
Problematize	5 minutes	Students work in teams to reflect on the day's investigations, discussing how their thinking has changed and what updates they plan to make to their SageModeler models in the next lesson. They

		prepare to apply their new insights to strengthen their models in the upcoming session.

MATERIALS

- Lesson 3.4 Slideshow
- Science Journal
- SageModeler Instruction Sheet (also in Student Journal, p. 51)
- SageModeler Ocean Food Web Preset Model

BEFORE YOU START TEACHING

1. For **Lesson 3.4**, review Student SageModeler Instructions (also in Science Journal, p. 51) to model or support students. Review SageModeler Ocean Food Web Preset Model.

2. Include a numbered list of general steps to set up the investigation before students arrive.

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: In this lesson, students investigate how rising atmospheric CO_2 changes the ocean's chemistry through ocean acidification. They will model how dissolved carbon dioxide forms carbonic acid and lowers ocean pH and investigate how this impacts organisms with calcium carbonate shells and the larger food web that depends on them. The lesson emphasizes systems thinking and cause-and-effect reasoning, helping students connect human carbon emissions to cascading biological and economic impacts.

Where We Are NOT Going: This lesson does not focus on the detailed chemical equations or quantitative analysis of acid-base reactions involved in ocean acidification.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

NAVIGATE: **Reflect** (10 minutes)

Lesson 3.3/3.4 Introduction Slides

1. Open the introduction slideshow to **Slide 1** and welcome students into the class and tell them that today, they will continue to learn about how carbon dioxide can change the rate of natural processes and affect the values that they care about.

2. Advance the slideshow to **Slide 2** and ask the class to take a moment and reflect on what they learned in lesson 3.2.
   - How did an increase of carbon dioxide emissions impact processes like global warming?
   - How do these changes affect the values your group placed on the ocean in your model?

   As students share, elevate ideas that mention: Increased levels of carbon dioxide emissions have an impact on the rate of change of global processes like global warming.

3. After student teams have reflected on the previous lesson, inform them that their next step is to find out what other processes might be impacted by increased carbon dioxide emissions.

4. Gather students in their teams from Module 2 and advance to **Slide 3**. Invite students to take a look at their team's model and brainstorm other processes that may be impacted by increased carbon dioxide emissions.
   - How may increased carbon dioxide emissions impact other processes that you have included in your team's model?

5. As students discuss with their teams, walk around and listen for conversations where you may support student brainstorming. Ask questions to guide students to choosing processes that we will investigate in lessons 3.3 and 3.4, sea level rise, coastal erosion, ocean acidification, and food web impacts.
   - What processes in your model have a large impact on the value your community cares about?
   - How might we investigate this process further?

6. After student teams have had a chance to review their models, invite the class back together to share some of their ideas.

7. Advance to **Slide 4** and tell students that for the next two sessions, they will embark on a student-led lesson, where they will learn more about particular processes that impact the value that their model centers on and ultimately update their team's model with their newfound information. They will have two options.
   - **Option 1:** Sea level rise and coastal erosion - You may choose this path if your model centers around the values of mental health and wellbeing, recreation, financial, and biodiversity.
   - **Option 2:** Ocean acidification and food web - You may choose this path if your model centers around the values of seafood, tradition, biodiversity, and cultural significance.

   To help guide students to choose their option, ask students the following question:
   - Of the two options provided, which processes might affect the value that centers your model the most?

8. When students have chosen their investigation, click on the links to lead students to these lessons.
   - Lesson 3.3 - Sea Level Rise and Coastal Erosion
   - Lesson 3.4 - Ocean Acidification

***DAY 1: NAVIGATE*:** **Exploring Ocean Acidification** (5 minutes)
Lesson plan notes are also provided on the slide notes.

Lesson 3.4 Ocean Acidification

1. Open the Ocean Acidification slideshow to **Slide 1,** where students will see a notice that this lesson is a student-led lesson, meaning that they will move through it with their team within the next two sessions.

2. Move on to **Slide 2,** where students will see the two-day overview for module 3.4.

3. When student teams are ready, advance to **Slide 3,** where they will see an image of a shell that is placed in an acidic solution (vinegar). Students are then asked to take 5 minutes to discuss with their group and answer the following questions.

    a. What do you think is happening in this image?
    b. Is what you are observing a natural occurrence?
    c. Why might communities want to understand this scenario?

4. Ask students to share their responses with the class. At this point, it is not important for them to be accurate with responses. You just want to gauge what they are noticing.

5. Advance to **Slide 4** and have students think back to their previous lessons in this module and ask:

    - What role do you think carbon dioxide might play in this process?
    - What happens when carbon dioxide interacts with ocean water?

6. After students have discussed, have them advance to **Slide 5,** where more information is shared with students about carbon dioxide interacting with ocean water.

The image shows a shell that has been placed in an acidic solution. Our ocean leans towards a more basic pH, but through reactions with atmospheric carbon dioxide, the ocean becomes slightly acidic. This is natural, but humans have impacted this reaction through the emissions of carbon dioxide.

7. Stay on **Slide 5,** where students will be told that over the next two lessons, they will investigate how this might happen through a process called ocean acidification. This includes how humans have accelerated the ocean acidification process and how this can impact coastal accessibility for communities.

DAY 1: INVESTIGATE: **Examining Ocean Acidification** (25 minutes)

1. Advance to **Slide 6** and let students know that to further explore the impacts that ocean acidification has on our oceans, they are going to set up an experiment.

2. On **Slide 6,** students will be introduced to a set-up for an experiment, where they will see how varying ocean pH levels affect organisms living in the ocean, just like the image they saw earlier.

3. Advance to **Slide 7** and have students set up the experiment following the experiment set-up instructions.

4. Once the experiment is set up, advance to **Slide 8** and ask students to turn to **Module 3.4 Ocean Acidification (Student Task, p. 46-49)** in their Science Journals. Here they will see an observation sheet for their experiment set-up.

5. On this observation sheet, students will see two modeling areas, one for the basic solution and one for the acidic solution. Have students take a moment to draw a model of what they think is happening in each of the solutions under the **"Start of Experiment"** section.

   As students are drawing a model, walk around and ask probing and pressing questions to support their thinking.

   - What do you think is happening in the solutions?
   - What processes or reactions are happening?
   - Are there any similarities between the two? Differences?

6. As students finish their initial model, advance to **Slide 9** and have them make a prediction on what they think will happen to the shell in each solution.
   - What do you predict will happen to the shells in each solution? Why?

   As students are making their predictions, walk around to answer questions they may have. If students need support making a prediction, provide a sentence frame to get

them started.
We predict that the shell in the [basic / acidic] solution will [what will happen] because [why you think it will happen]

7. After students have made their predictions, advance to **Slide 10**, where students will have instructions to store their experiment for day two.

DAY 1: NAVIGATE: Sharing What We've Learned (5 minutes)

1. As you reach the end of the class period, bring the students back together for a class conversation (See Module 3.3 & 3.4 Introduction Slides, Slide 5). As not all students did the exact same activity, get at least one student from each exploration to share responses to the following questions with the class.

    - What did you do today?
    - What were some of the most surprising findings you gained from today?

2. As students share their experiences, ask the following questions:

    - What specific data do you think is still missing to further your understanding?
        - What might help strengthen your understanding of carbon dioxide's impact on sea level rise?
        - What might help strengthen your understanding of carbon dioxide's impact on ocean acidification?
    - Where might you collect this data?

DAY 1: PROBLEMATIZE: Reflecting on the Day (5 minutes)

1. As students wrap up for the day, send groups back to their slideshow to reflect on the process of learning about ocean acidification and creating their experiment models (Lesson 3.4, slide 10).

2. Advance to **Slide 11** and have student teams work together to answer the following questions together on Day 1: **Module 3.4 Ocean Acidification (Student Reflection)**, p. 50 in their Science Journal.
    - What new information or evidence changed your thinking about how CO_2 affects ocean ecosystems?
    - What questions do you still have about ocean acidification?

3. Thank students for their participation today and inform them that in the next session, they will continue to learn more about ocean acidification!

DAY 2: NAVIGATE (10 minutes): Further Exploring Ocean Acidification
Lesson plan notes are also provided on the slide notes.

1. Welcome the students back for day two of their Ocean Acidification investigation. Have students open the slideshow to **Slide 12,** where day two starts.

2. When students are ready, tell them that the first task they have is to watch a quick video to further learn about Ocean Acidification. Advance to **Slide 13,** where students will play the video, What is Ocean Acidification? [2:05].

3. After the students have watched the video, advance to **Slide 14,** where they will see the same image of the shell in an acidic solution from day one to see how their thinking has changed. As a group, have students answer the following questions.
    - What do you think is happening in this image? Why is it happening?
    - Is what you are observing a natural occurrence?
    - Why might communities want to understand this scenario?

4. Have students share with their group. When they are ready, they can advance to **Slide 15** where they will see their goal for today.
    - Today, you will further explore your experiment and use the information you gather to plan how you want to update your SageModeler model to describe how ocean acidification impacts the values your community places on Southern California's ocean.

DAY 2: INVESTIGATE: **Exploring Our Predictions** (25 minutes)
1. When students are ready, advance to **Slide 16** and have students open their Science Journal to **Module 3.4 Ocean Acidification (Student Task) p. 46-49,** where they made observations in the previous lesson**.**

2. Have students bring out their experiments from day one and make observations on how they have changed, drawing and labeling their observations in the section that reads

"Day Two of Experiment".

As students are drawing their observations, go around and ask probing and pressing questions.

- What do you notice? How is this different from the beginning of the experiment?
- What processes or reactions happened? What do you think caused this to happen?
- How are the two experiments different? Similar?

3. Once students have finished filling out their observation sheets, have them clean up the experiments and ask them to move to **Module 3.4: Impact of Carbon Dioxide Emissions on the Ocean's pH and food web (*Day 2: Sage Modeler Instructions)***, p. 51 in their Science Journals. Here, they will be asked to think about how to apply all the information they have gathered through exploring a pre-set SageModeler model.

4. Advance to **Slide 17,** where students will see a link to visit a preset SageModeler model. Here, they will be asked to think about how to apply all the information they have gathered.

5. Open the SageModeler model and ask the students to follow along with their Science Journal on **Module 3.4: Impact of Carbon Dioxide Emissions on the Ocean's pH and Food Web (*Day 2: Sage Modeler Instructions)*, p. 51.** As students are investigating the model, walk around and provide support by asking probing questions.

   - How does carbon dioxide get introduced into the Earth's system in the model?
   - What happens to the seawater as it comes in contact with carbon dioxide?
   - How can you manipulate this model to reflect the basic solution and acidic solution scenarios you observed during your experiment?
   - How does increasing the amount of carbon dioxide impact the living organisms in this model?

6. As students are finishing up, advance to **Slide 18** and ask them.

   - How does a change to the ocean's pH level affect the value your community places on Southern California's ocean?

***DAY 2: NAVIGATE*: Discuss How to Update our Models** (10 minutes)

1. As students wrap up their investigation, advance to **Slide 19** and tell students that their next step is to create a plan to update their SageModeler model from Module 2 with their newfound knowledge during their investigation.

2. Have students open their Science Journal in Module 3.4: Ocean Acidifications (*Planning to Update Our SageModler Model*) (p.54). On this page, they will have a table to help

them think through their updates. As students work together, walk around the groups and support them in answering the questions.

- What investigation did you take part in?
- How did your thinking change as a result of this investigation? What new information did you learn?
- What do you want to add, remove, or update in your SageModeler model?

  Students may need support in thinking about how to expand their original model. Invite students to take a look back at their materials from module 3 so far and look at the components they investigated.
  Some options may include
  - **For all investigations:** carbon dioxide emissions and atmospheric carbon dioxide.
  - **For sea level rise and coastal erosion investigations:** melting land ice, shoreline width, or flooding
  - **For ocean acidification investigations:** ocean pH levels, organisms with calcium carbonate shells, or connections back to the food chain that impact their value, for example, the availability of fish.

3. Once students have created their plan, bring the class back together (Reference AISciComm Module 3.3 & 3.4 Introduction Slides, slide 7) and have them share out with the class if time permits. Since different students will have different investigations and their models share different values, focus on the following question.

   - What value does your SageModeler model center on?
   - What investigation did you take part in? (sea level rise or ocean acidification)?
   - How do you want to update your SageModeler model as a result of your investigation? How does this impact connect with the value your model centers?
4. Then, return to AISciComm Module 3.4 Slides, slide 20

DAY 2: PROBLEMATIZE: **Reflecting on the Day** (5 minutes)

1. As students wrap up for the day, their final task for the day is to reflect on what they did.

2. Advance to **Slide 20** and have student teams work together to answer the following questions together in **Module 3.4 Ocean Acidification (Day 2: Student Reflection), p.55** of their Science Journals.
   - How does ocean acidification affect the ocean food web and the people who depend on it?
   - What questions do you still have about ocean acidification?
   - What updates do you plan to make to your SageModeler in the next lesson to strengthen your model?

3. Thank students for their participation today and inform them that in the next session, they will come back to their SageModeler model and update it with their newfound knowledge from their investigations.